

ALGORITHM
Spring 2023, Midterm Exam, April 13
5:00 – 8:00 pm

1. (20%) Illustrate the operation of the following sorting algorithms respectively on the array $A = \langle 8, 5, 1, 4, 7, 6, 4, 2, 1, 3, 6 \rangle$, where $A[j] \in \{0, 1, \dots, 8\}$ for $1 \leq j \leq 11$.
Which of them are stable sorting algorithms? Which of them are in-place sorting algorithms?
 - (a) MERGE SORT
 - (b) BUBBLE SORT
 - (c) INSERTION SORT
 - (d) QUICKSORT
 - (e) COUNTING SORT
2. (30%) Please answer the following questions.
 - (a) Show that for each even number n greater than 4, there exists a 5-regular graph $G(V, E)$ with n nodes.
 - (b) Illustrate the operation of MAX-HEAP-INSERT($A, 3$) on the heap $A = \langle 13, 12, 9, 10, 12, 8, 7, 4, 5, 6, 2, 1 \rangle$.
 - (c) Use the master method to give a tight asymptotic bound on the running time of the recurrence $T(n) = 4T(n/2) + n^4$.
 - (d) Show that an n -element heap has height $\lfloor \lg n \rfloor$.
 - (e) Show that the running time of QUICKSORT is $\Theta(n^2)$ when the array A contains distinct elements and is sorted in decreasing order.
 - (f) Which of them are in-place sorting algorithms? a : INSERTION SORT b : MERGE SORT
 c : QUICKSORT d : HEAPSORT
3. (20%) Let $f(n)$ and $g(n)$ be real-valued functions. The asymptotic notations, O , Ω , and Θ , are defined as follows.
 - $f(n)$ is *asymptotically upper bounded by* $g(n)$, denoted by $f(n) = O(g(n))$,
iff there exists positive constants c and n_0 such that $f(n) \leq c \cdot g(n)$ for all $n > n_0$.
 - $f(n)$ is *asymptotically lower bounded by* $g(n)$, denoted by $f(n) = \Omega(g(n))$,
iff there exists positive constants c and n_0 such that $f(n) \geq c \cdot g(n)$ for all $n > n_0$.
 - $f(n)$ is *asymptotically equivalent to* $g(n)$, denoted by $f(n) = \Theta(g(n))$,
iff there exists positive constants c_1, c_2 and n_0 such that $c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$
whenever $n > n_0$.

Please answer the following questions according to the definitions given above.

- (a) Finding upper bound for the function $f(n) = n^3 + 3n - 2$ by using the O notation.
Justify your answer by demonstrating the constants.
- (b) Finding lower bound for the function $f(n) = n^3 + 3n - 2$ by using the Ω notation.
Justify your answer by demonstrating the constants.

- (c) Finding tight bound for the function $f(n) = n^3 + 3n - 2$ by using the Θ notation. Justify your answer by demonstrating the constants.
- (d) Show that $f(n) = n^3 + 3n - 2 \neq \Omega(n^4)$.
4. (15%) Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 4T(\lfloor n/4 \rfloor) + 2n$, where $T(1) = 2$, $T(2) = 5$ and $T(3) = 8$. Use the substitution method to verify your answer.
5. (15%) Given an array $A = \langle x_1, x_2, \dots, x_n \rangle$ of numbers, the maximum-subarray problem is to find two indices i and j such that $A = \langle x_i, x_2, \dots, x_j \rangle$ has the greatest sum of any contiguous subarray of array A .
- (a) Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -3, 5, -3, 4, -2, 2, 4, -3 \rangle$.
- (b) Describe an algorithm that solves the maximum-subarray problem in $O(n \log n)$ time.
6. (15%) Given an ordering of preferences for each element, the stable matching problem is to find a stable matching between two equally sized sets of elements. A matching is stable when no two unmatched partners prefer each other over their current partners.
- (a) Is (周, 昆), (錦, 劉), (梁, 蔡) a stable matching?
- (b) Describe an algorithm that solves the stable matching problem, and analyze its asymptotic worst-case running time.

	1 st	2 nd	3 rd
周杰倫	蔡依林	昆凌	劉嘉玲
錦榮	蔡依林	昆凌	劉嘉玲
梁朝偉	昆凌	蔡依林	劉嘉玲

	1 st	2 nd	3 rd
蔡依林	錦榮	周杰倫	梁朝偉
昆凌	周杰倫	錦榮	梁朝偉
劉嘉玲	周杰倫	錦榮	梁朝偉

7. (15%) Famous ride-sharing company NTOU has created a new “shared-bus” service. The bus serves a path of m miles, indexed by $[0, m]$. The main differences between this service and traditional bus services are:
- All the passengers board the bus at point 0.
 - Passenger i specifies a range $[s_i, t_i]$, where $0 \leq s_i \leq t_i \leq m$. The passenger is satisfied as long as the bus stops at some location between s_i and t_i (including the boundary).

There are no fixed bus stops. Instead, the stops will be decided after gathering the user information. As an engineer at NTOU, you need to solve the following problem: given the number of passengers n , and each passenger's preference $[s_i, t_i]$, design an algorithm that finds the smallest number of stops in order to satisfy all customers.

- (a) Design the algorithm and analyze its running time. Your algorithm should run in $O(n \log n)$ time.
- (b) Prove the correctness of the algorithm you designed in (a).